5/26/26

PHY202 Exam 1 on chapter(s): 16, 17

100% Excellent!

Name:_

These questions assess your understanding of the material from Chapter(s) 16, 17 of OpenStax College Physics. Please write your name on the sheets of paper that you use. Carefully read each question. Credit is only awarded if you answer correctly and clearly show all major steps necessary to find your answer; credit is not awarded for missing work, unclear work, or the use of memorized equations absent from the equation sheet. Half credit may be awarded for insufficient work that suggests some degree of understanding. Half credit is awarded for correct numerical answers with incorrect or missing units. Correct numerical answers do not guarantee credit.

1. When Ethan gets in his car, it lowers by 0.51 cm. If Ethan's mass is 95.0 kg, then what is the force constant of his car's suspension system?

$\vec{F} = -k\Delta\vec{x} \rightarrow F = -k\Delta x = mg \rightarrow k = \frac{mg}{-\Delta x}$

ANSWER: $1.83\times10^5\ \frac{N}{m}$

$k = \frac{(95\ kg)(9.8\ \frac{m}{s^2})}{-(-0.0051\ m)} = \boxed{182549\ \frac{kg}{s^2}}$

2. Suppose you have a tube that is closed at one end. Its length is 8.3 cm and the air temperature is 0.24°C. What is the frequency of its first overtone?

$n=3$

ANSWER: 2993 Hz

$f_n = n\frac{v_w}{4L} = \frac{n(331\ \frac{m}{s})\sqrt{\frac{T}{273\ K}}}{4L} = \frac{(3)(331\ \frac{m}{s})\sqrt{\frac{(0.24+273.15)K}{273\ K}}}{4(0.083\ m)} = \boxed{2993.099\ \frac{1}{s}}$

3. If the shock absorbers in a car go bad, then the car will oscillate at the smallest provocation, such as when going over bumps in the road and after stopping. Calculate the period of oscillations for a car with shoddy shock absorbers if the car's total mass is 845. kg and the force constant of the suspension system is $1.01\times10^4\ \frac{N}{m}$.

ANSWER: 1.82 s

$T_{SHO} = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{845\ kg}{1.01\times10^4\ \frac{kg}{s^2}}} = \boxed{1.817\ s}$

+3

4. Suppose a train sounding its 144.-Hz horn is moving at 34.3 $\frac{m}{s}$. What frequency does a person standing by the tracks hear as the train approaches? Given this particular day's temperature, the speed of sound is 331. $\frac{m}{s}$.

ANSWER: 161 Hz

$$f_{obs} = f_s\left(\frac{v_w}{v_w - v_s}\right) = (144\ Hz)\left(\frac{331\ \frac{m}{s}}{331\ \frac{m}{s} - 34.3\ \frac{m}{s}}\right) = \boxed{160.647\ Hz}$$

5. Suppose you have a tube that is closed at one end. If you want its fundamental frequency to be 126. Hz and the air temperature is 0.16°C, then what should the length of the tube be?

ANSWER: 0.657 m

$n=1$

$$f_n = n\frac{v_w}{4L} \rightarrow 4Lf_n = nv_w \rightarrow L = \frac{nv_w}{4f_n} = \frac{n(331\ \frac{m}{s})\sqrt{\frac{T}{273\ K}}}{4f_n}$$

$$L = \frac{1(331\ \frac{m}{s})\sqrt{\frac{(0.16+273.15)K}{273\ K}}}{4(126\ \frac{1}{s})} = \boxed{0.657\ m}$$

6. The spring of a toy gun has a force constant of 92.8 $\frac{N}{m}$ and is compressed 9.46 cm. Neglecting friction and the mass of the spring, at what speed will a 2.40-g projectile be ejected from the toy gun?

ANSWER: 18.6 $\frac{m}{s}$

$$\Delta E_t = W_{nc} = 0 = \Delta KE + \Delta PE = KE_f - \cancel{KE_i} + \cancel{PE_f} - PE_i \rightarrow KE_f = PE_i = \frac{1}{2}mv_f^2 = \frac{1}{2}kx_i^2$$

$$v_f = \sqrt{\frac{kx_i^2}{m}} = \sqrt{\frac{(92.8\ \frac{kg}{s^2})(0.0946\ m)^2}{0.0024\ kg}} = \boxed{18.602\ \frac{m}{s}}$$

+3

7. Suppose that a 1.30×10^3 -kg car has a suspension system with a force constant $3.57 \times 10^4 \frac{N}{m}$. If the car hits a bump and bounces with an amplitude of 6.6 cm, then what is its maximum vertical velocity assuming no damping occurs?

ANSWER: 0.346 m/s ✓

$$v_{max} = \sqrt{\frac{k}{m}} X \boxed{\sin\left(\frac{2\pi t}{T}\right)}^{-1} = (0.066\ m)\sqrt{\frac{3.57\times10^4\ \frac{kg}{s^2}}{1.3\times10^3\ kg}} = \boxed{0.3459\ \frac{m}{s}}$$

8. A medical imaging device produces ultrasound by oscillating with a period of 0.12 μs. What is the frequency of this oscillation?

ANSWER: 8.3×10^6 Hz ✓

$$f = \frac{1}{T} = \frac{1}{0.12\times10^{-6}\ s} = \boxed{8333333.33\ \frac{1}{s}}$$

9. Ultrasound is transmitted from fat to muscle during a particular medical procedure. Calculate the percentage of the ultrasound's intensity that is reflected when the ultrasound travels from fat with a density of 856. $\frac{kg}{m^3}$ to muscle tissue with a density of $1.002 \times 10^3\ \frac{kg}{m^3}$. The speed of ultrasound in the fat is $1.50 \times 10^3\ \frac{m}{s}$ and the speed of ultrasound in the muscle tissue is $1.58 \times 10^3\ \frac{m}{s}$.

ANSWER: 1.09% ✓

Z_1 = Fat, Z_2 = muscle

$$a = \frac{(Z_2-Z_1)^2}{(Z_1+Z_2)^2} = \frac{(\rho_2 v_2 - \rho_1 v_1)^2}{(\rho_1 v_1 + \rho_2 v_2)^2} = \frac{\left((1.002\times10^3\ \frac{kg}{m^3})(1.58\times10^3\ \frac{m}{s}) - (856\ \frac{kg}{m^3})(1.5\times10^3\ \frac{m}{s})\right)^2}{\left((856\ \frac{kg}{m^3})(1.5\times10^3\ \frac{m}{s}) + (1.002\times10^3\ \frac{kg}{m^3})(1.58\times10^3\ \frac{m}{s})\right)^2}$$

$$a = 0.0109 = \boxed{1.09\%}$$

+3

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1. $k = 1.83 \times 10^5\ \frac{\mathrm{N}}{\mathrm{m}} \because \vec{F} = -k\Delta\vec{x}.\ k = -\frac{\vec{F}}{\Delta\vec{x}}$
(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex1)

2. $f = 2.99 \times 10^3\ \mathrm{Hz} \because f_n = n\frac{v_w}{4L},\ v_w = \left(331\,\frac{\mathrm{m}}{\mathrm{s}}\right)\sqrt{\frac{T}{273\ \mathrm{K}}}$
(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5b)

3. $f = 1.82\ \mathrm{s} \because T_{SHO} = 2\pi\sqrt{\frac{m}{k}}$
(Chapter 16, Section (3) Simple Harmonic Motion - cp1e_ch16_ex4)

4. $f = 160.6\ \mathrm{Hz} \because f_{obs} = f_s\left(\frac{v_w}{v_w \pm v_s}\right)$ (review the Doppler effect)
(Chapter 17, Section (4) Doppler Effect and Sonic Booms - cp1e_ch17_ex4)

5. $L = 0.6569\ \mathrm{m} \because f_n = n\frac{v_w}{4L},\ v_w = \left(331\,\frac{\mathrm{m}}{\mathrm{s}}\right)\sqrt{\frac{T}{273\ \mathrm{K}}}$
(Chapter 17, Section (5) Sound Interference and Resonance: Standing Waves in Air Columns - cp1e_ch17_ex5a)

6. $v = 18.6\ \frac{\mathrm{m}}{\mathrm{s}} \because KE = PE_{elastic}$
(Chapter 16, Section (1) Hooke's Law - cp1e_ch16_ex2)

7. $v_{max} = 0.346\ \frac{\mathrm{m}}{\mathrm{s}} \because v(t) = \sqrt{\frac{k}{m}}X\sin\frac{2\pi t}{T}$ and the max of sine's range is 1.
(Chapter 16, Section (5) Energy and the Simple Harmonic Oscillator - cp1e_ch16_ex6)

8. $f = 8.33 \times 10^6\ \mathrm{Hz} \because T = \frac{1}{f}$
(Chapter 16, Section (2) Period and Frequency in Oscillations - cp1e_ch16_ex3)

9. $a = 1.09\% \because Z = \rho v,\ a = \frac{(Z_2 - Z_1)^2}{(Z_1 + Z_2)^2}$
(Chapter 17, Section (7) Ultrasound - cp1e_ch17_ex7)